NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Data Science Fundamentals [SC3021]

Chapter 4: Data Preparation

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Chapter content

Data science overview

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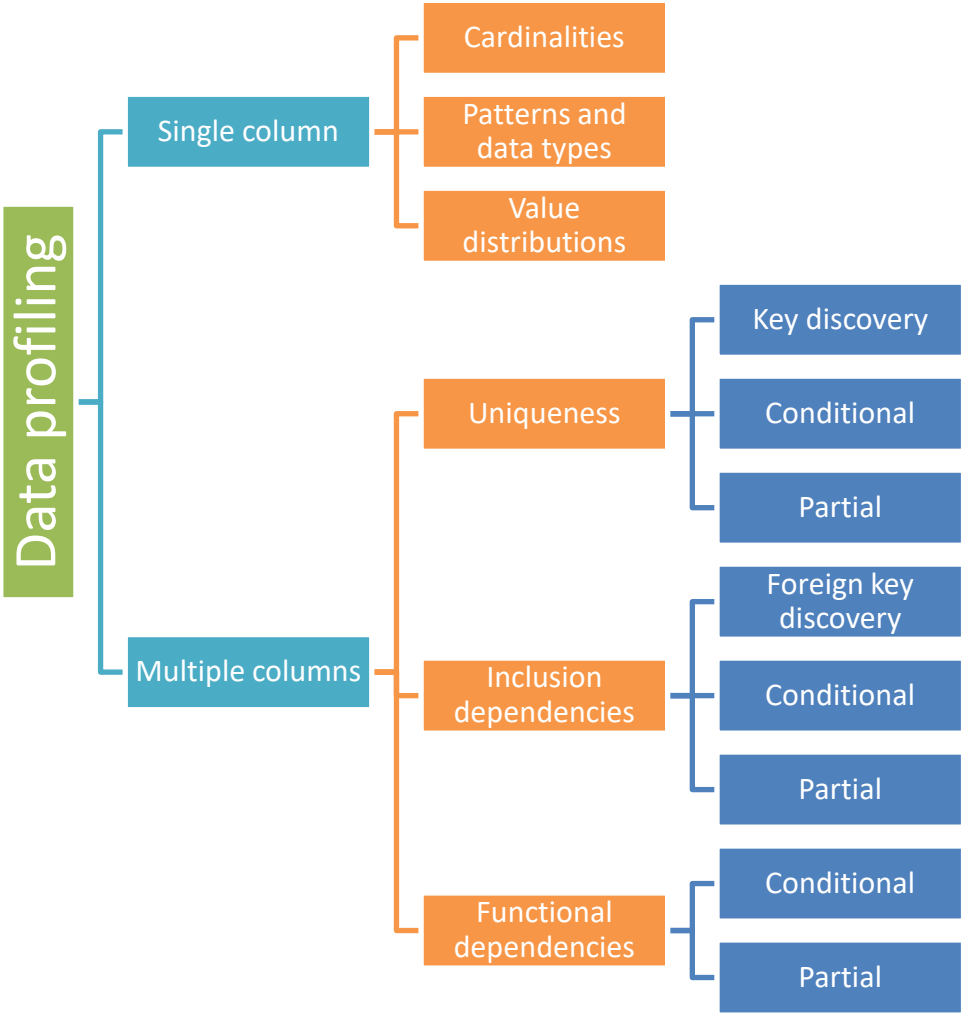
Chapter 4.2: Data profiling (continued)

Classification of standard profiling tasks

- Applies to structured (tabular) data.
- Distinguishes between profiling of data stored in a single column and data stored across multiple columns.
- For profiling tasks on multiple columns, exact, conditional, and partial tasks exist.



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Single-column vs. multi-column

Single column profiling

- Most basic form of data profiling
- Assumption: All values are of same type
- Assumption: All values have some common properties that are to be discovered
- Often part of the basic statistics gathered by DBMS
- Complexity: Number of values/rows

Multi-column profiling

- Discover joint properties
- Discover dependencies
- Complexity: Number of columns and number of values

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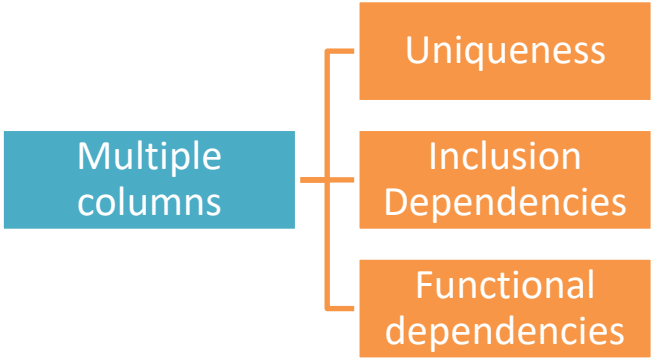
Single-column vs. multi-column



Multi-column profiling

- Discover joint properties
- Discover dependencies
- Complexity: Number of columns and number of values

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Uniqueness and keys

- **Unique column:** all values are unique
- **Unique column combination**
  - Considers values from multiple attributes A
  - Only unique value combinations when combining values of A
  - Minimality: No strict subset $A' \subset A$ is a unique column (combination)
- **Key candidate:** Unique values AND no null values
- **Key**
  - Column / column combination that serves as general unique identifier for entities modeled by a table.
  - Uniqueness and non-null in one instance of a table does not imply key: Only human can specify keys.
- **Applications**: schema design, data integration, indexing, query optimization.

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Uniqueness and keys

Name

Text

Unique

Distinct	891
Distinct (%)	100.0%
Missing	0
Missing (%)	0.0%
Memory size	73.2 KiB

Titanic dataset exploration with ydata

Unique column

No missing values

=> Key candidate

=> Not a key, because, in general, it is possible that multiple passengers boarding a ship have the same name.

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Uniqueness and Keys

Firstname	Lastname	Age	ZIP
John	Doe	42	1234
Jane	Doe	39	9876
Jane	Smith	42	2468
John	Doe	22	1234

No single column is unique

Column combination (Firstname, Lastname, Age, ZIP) is unique, BUT not minimal!

Minimal unique column combinations:
(Lastname, Age)
(Firstname, Age)
(Age, ZIP)

Discovery algorithms find all, relevance assessed by domain expert.

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Inclusion dependencies and foreign keys

- $A \subseteq B$: All values in attribute A are also present in attribute B
- $A_1,\ldots,A_i \subseteq B_1,\ldots,B_i$:
  All value combinations in $A_1,\ldots,A_i$ are also present in $B_1,\ldots,B_i$
- Prerequisite for foreign key that defines attributes referencing attributes of another table (via value equality)
  - Used across relations
  - Use across databases
  - But again: Discovery on a given instance, only user can specify if it holds in general and should be defined as foreign key at schema level.
- **Applications**: schema design, data integration, indexing, query optimization.

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Inclusion dependencies and foreign keys

Data from one source S1

Firstname	Lastname	Age	ZIP
John	Doe	42	1234
Jane	Doe	39	9876
John	Smith	42	2468
John	Doe	22	1234

Data from another source S2

ZIP	City
1234	Blue Mountain
2345	Green Forest
2468	Big River
5588	Gold Hill
9876	Ice Lake

$S1.ZIP \subseteq S2.ZIP$

but not vice versa

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Functional dependencies

- Let X be a set of attributes and A be a single attribute.
- A functional dependency exists from X to A if, whenever two records have the same X values, they also have the same A values.
- A functional dependency is written as $X \rightarrow A$ (pronounced X determines A).
- **Applications**: schema design, data cleaning

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Functional dependencies

Firstname	Lastname	Age	ZIP	City
John	Doe	42	1234	Blue Mountain
Jane	Doe	39	9876	Ice Lake
John	Smith	42	2468	Big River
John	Doe	22	1234	Blue Mountain
Jack	Smith	65	9876	Ice Lake
Jill	Park	22	2469	Big River

ZIP → City holds

But City does NOT determine ZIP (one city can have several zip codes)

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Partial dependencies

- Also known as “approximate dependencies”
- Refers to dependencies that do not perfectly hold.
- Examples: Dependency holds
  - For all but 10 of the tuples
  - Only for 80% of the tuples
  - Only for 1% of the tuples
- Classification of data profiling tasks includes partial dependencies for multi-attribute profiling tasks.
- Generalizes to further profiling tasks, e.g., for patterns, types, etc.
- **Applications**: data quality assessment, data cleaning

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Partial dependencies

Firstname	Lastname	Age	ZIP	City
John	Doe	42	1234	Blue Mountain
Jane	Doe	39	9876	Ice Lake
John	Smith	42	2468	Big River
John	Doe	22	1234	BM
Jack	Smith	65	9876	Ice Lake
Jill	Park	22	2469	Big River

ZIP → City **almost** holds

Data quality issue: City name was abbreviated, which is not the standard format.

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Conditional dependencies

- Given a partial inclusion dependency or functional dependency, conditional dependencies define for **which** part of the data they hold.
- Expressed as a condition over the attributes of the relation.
- Problems:
  - Infinite possibilities of conditions
  - Interestingness:
    - Many distinct values: less interesting
    - Few distinct values: surprising condition – high coverage
- **Applications**: data integration, data cleaning

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Conditional dependencies

Data from one source S1 covering several countries

Firstname	Lastname	Age	ZIP	Country
John	Doe	42	1234	E
Jane	Doe	39	9876	E
John	Smith	42	2468	E
John	Doe	22	1234	E
Pierre	Dupont	58	1234	F
Jeanne	Petit	63	5712	F

Data from another source S2 covering country E only

ZIP	City
1234	Blue Mountain
2345	Green Forest
2468	Big River
5588	Gold Hill
9876	Ice Lake

S1.ZIP ⊆ S2.ZIP only partially holds.

It holds (exactly) for tuples where S1.Country = E

→ This conditional inclusion dependency helps us to correctly integrate data.

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Computational complexity

- Identifying multi-column constraints may require to consider all possible column combinations
  → **exponential** in the number of columns
- Checking if a constraint holds for one specific column combination requires checking the data, e.g., by comparing all rows to each other → **quadratic** in the number of rows
- Altogether, this is considered **practically infeasible for large amounts of data**.
- **Designing efficient algorithms** for multi-column profiling is important.

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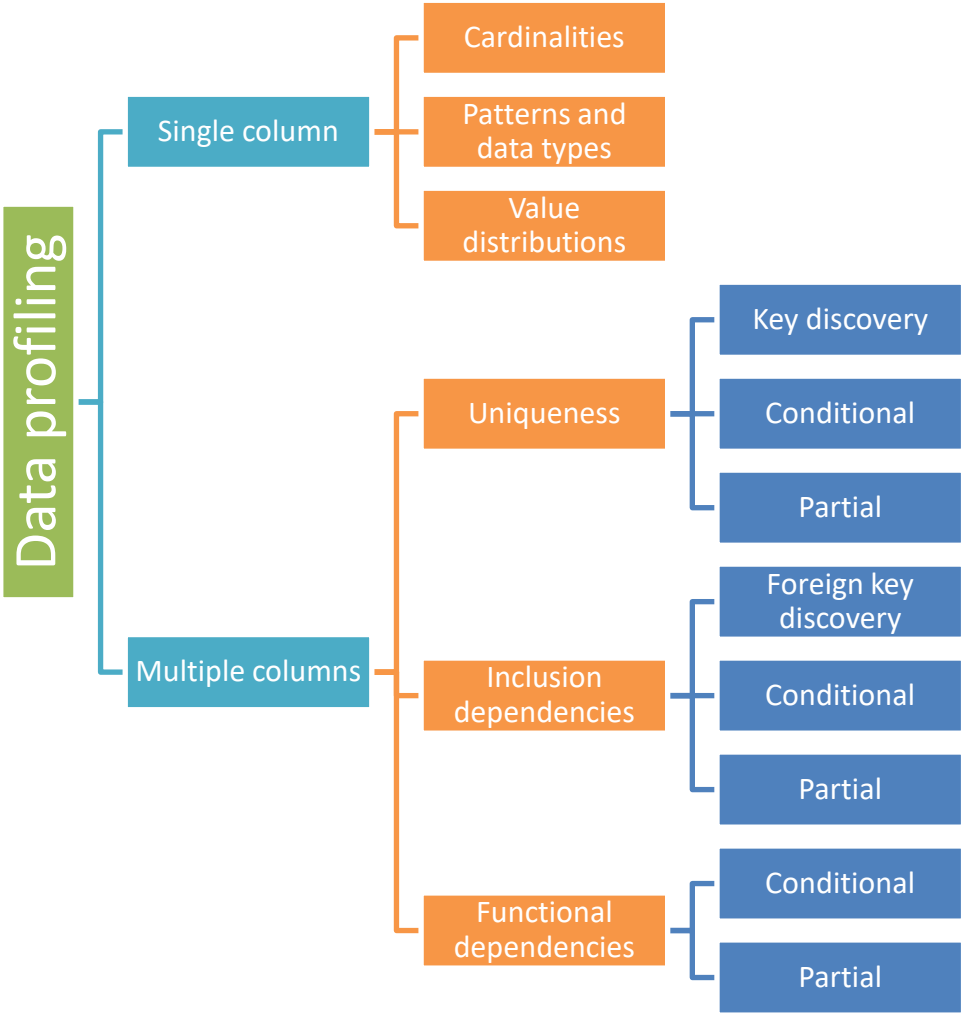
Data profiling classification summary

- Distinguishes between profiling of data stored in a single column and data stored across multiple columns.
- For profiling tasks on multiple columns, exact, conditional, and partial tasks exist.
- Applications vary for different profiling tasks, but all are relevant to data science.



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CONCLUSION

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Summary

- Data evolves through stages (from raw over refined to production)
- Data preparation is essential for data actions and analytical actions across all data flow stages.
- Two critical tasks of data preparation are data profiling and data transformation.
- Data profiling is the set of activities and processes to determine relevant metadata about a given dataset.
- Such metadata can relate to single columns or multiple columns.
- Data profiling applies at various stages and for various applications throughout data processing.

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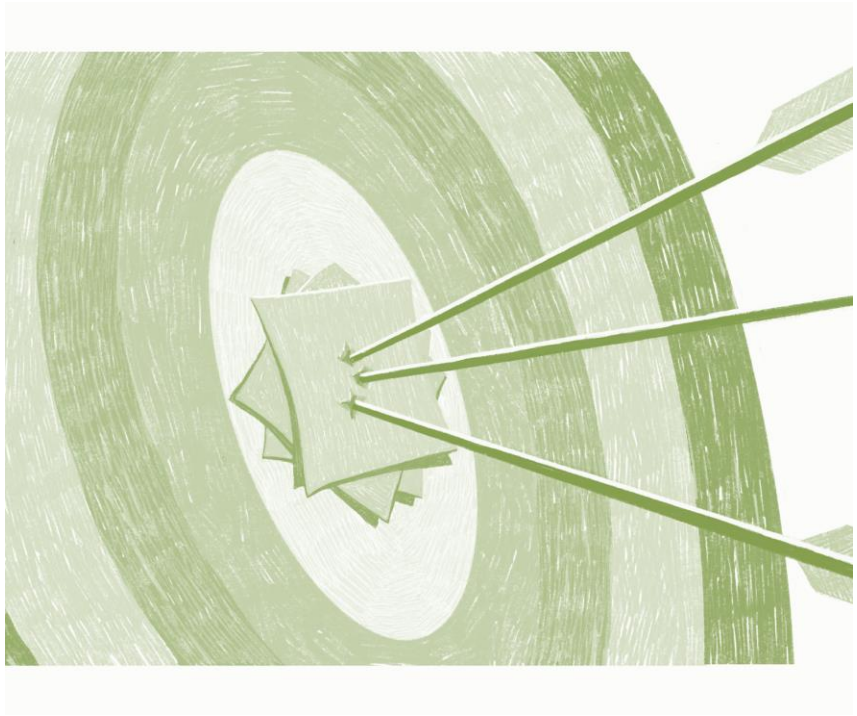
You should now be able to…

- … explain different profiling tasks and how they may apply at different data science stages.
- … identify profiling tasks suited to gain relevant information on a given dataset for a specific application.
- … reflect on the computational complexity of profiling tasks when applying them to large data sets.



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References and credits

- Joseph M. Hellerstein, Tye Rattenbury, Jeffrey Heer, Sean Kandel, Connor Carreras. Principles of Data Wrangling. O'Reilly Media. 2017 – Chapter 4
- Ziawasch Abedjan, Lukasz Golab, Felix Naumann: Profiling relational data: a survey. VLDB J. 24(4): 557-581 (2015)

- ***Slides partially based on slides shared by Felix Naumann, Hasso-Plattner-Institut Potsdam, Germany***

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